Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Genetos, Damian C Ph.D.		
Project Title:	Activation, Adaptation, Restoration of the Mechanical Response of Osteoblasts to Fluid Shear		
Division Name:	Human Research		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	717-531-6697
Organization Name:	Penn State College of Medicine		
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PI Web Page:			
City:	Hershey	**State:**	PA
Zip Code:	17033	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2001	**End Date:**	06/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A primary biomedical concern during extended spaceflight is the rapid decrease in bone mass associated with microgravity. Conversely, application of mechanical stimulation produces osteogenesis. Osteoblasts respond to mechanical stimulation with a rapid increase in intracellular Ca2+ that we postulate is required for this osteogenic response. In this application, we propose to examine the role of mechanosensitive (MSCC) and L-type voltage-sensitive (VSCC) Ca2+ channels and intracellular Ca2+ release in the activation, habituation (or adaptation) and restoration of mechanosensitivity in osteoblasts in response to fluid shear.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	For bone architecture, form follows function. Decreased function increases osteoclastic resoprtion, and weakening, of bone. Mechanically loading bone increases formation and mechanical strength, although this process is saturated after a certain time. By determining the molecular mechanisms involved in the saturation of a response, it will be possible to better understand optimal loading conditions to create an anabolic response of bone.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 08/28/2017)
Articles in Peer-reviewed Journals	Liu D, Genetos DC, Shao Y, Geist DJ, Li J, Ke HZ, Turner CH, Duncan RL. "Activation of extracellular-signal regulated kinase (ERK1/2) by fluid shear is Ca(2+)- and ATP-dependent in MC3T3-E1 osteoblasts." Bone. 2008 Apr;42(4):644-52. Epub 2007 Oct 16. https://doi.org/10.1016/j.bone.2007.09.058 ; PubMed PMID: 18291742; PubMed Central PMCID: PMC2937351 , Apr-2008